Creative thinking and problem solving

Creative thinking is a divergent process, using the imagination to create a large range of ideas for solutions. It requires us to look beyond the obvious, creating ideas which may, at first, seem unrealistic or have no logical connection with the problem. There is a large element of creative thinking in solving open problems. The creative thinking skills can be divided into several key elements:

fluency - producing many ideas

flexibility - producing a broad range of ideas

originality - producing uncommon ideas

elaboration - developing or expounding ideas.

Effective problem solving requires a controlled mixture of analytical and creative thinking. To be a good problem solver you need to be able to switch from one group of skills to the other and back again, although this is not always easy.

Some other terms which are often used in discussions of creativity include:

Intuition - the ability to draw conclusions based on impressions and feelings rather than hard facts. It is a characteristic of right-brain thinking and some people rely on it more than others.

Incubation - the period between stopping conscious work on a problem and the time when we become aware of a solution or part of solution. People struggling with problems often suddenly become aware of a solution after a period of incubation, during which the mind is occupied by other things.

Invention - the creation of new, meaningful ideas or concepts.

Innovation - putting new ideas or concepts to a practical use, as in the development of a new product or service.

Brainstorming

Brainstorming is a popular technique for generating creative solutions for complex problems. It involves generating as many ideas as possible without judging, evaluating, or censoring them. The goal is to explore different perspectives, possibilities, and alternatives, and to stimulate your

creative thinking skills. In this article, you will learn how to use brainstorming effectively in six steps.

1 Define the problem

Before you start brainstorming, you need to have a clear and specific definition of the problem you want to solve. This will help you focus your ideas and avoid confusion or ambiguity. You can use a problem statement, a question, or a goal to define the problem. For example, "How can we increase customer satisfaction and loyalty?" or "What are some ways to reduce our environmental impact?" One of the really important aspects of defining the problem is to do so without leading the participants towards a solution. Keeping the brainstorming exercise as open and unconstrained as possible will allow for a wider range of potential solutions.

2 Set the rules

Brainstorming is most successful when certain rules are followed to encourage creativity and participation. These rules include avoiding criticism or evaluation of ideas, striving for quantity over quality, building on other ideas, and being open and supportive. Everyone should listen to one another's ideas, respect their opinions, and avoid dominating or interrupting the conversation. Generating numerous ideas that don't worry about feasibility, practicality, or originality will help create the best results.

The important thing about setting the rules is that you clearly define your expectations for the exercise both in what the participant should expect (how to share ideas, how to interact with other participants, how long the session will last, how ideas will be evaluated) as well as what your desired outcome for the exercise is.

3 Choose a method

There are various methods of brainstorming that you can choose from depending on your preferences, objectives, and resources. Brainwriting is a written form of brainstorming which involves each person writing down their ideas on a piece of paper or a sticky note and passing it to the next person for further input. Mind mapping is a visual approach to brainstorming where you start with a central idea or problem and branch out into related subtopics, concepts, or keywords. Role playing is another creative way of brainstorming which involves pretending to

be someone else who has different perspectives, experiences, or expertise on the problem. This can be done through characters, scenarios, or questions to stimulate your imagination and empathy.

This might need to be moved up in the order to "Set the Rules". The methodology that is chosen will impact how it is presented to the participants. It is important to note that in the methodology portion of Brainstorming that it is important to visibly record all ideas in a way that the participants can see them. For readability, consider changing the presentation of the examples (Brainwriting, Mind Mapping, and Role Playing) to bullets. Might also want to add additional examples.

4 Generate ideas

Once you have chosen a method, you can start generating ideas. You can brainstorm individually or in a group, depending on the situation and the availability of other people. If you brainstorm in a group, you can assign a facilitator, a recorder, and a timekeeper to help manage the process. The facilitator guides the discussion, encourages participation, and enforces the rules. The recorder captures the ideas on a board, a paper, or a digital tool. The timekeeper monitors the time and signals when to start and stop.

Incorporate creativity techniques like mind mapping, analogies, role-playing, or reverse thinking to stimulate innovative ideas. These techniques can help break through mental barriers and uncover new possibilities. It's important to foster an atmosphere of open communication and non-judgment. All ideas should be accepted and recorded without immediate evaluation or criticism. This encourages participants to think creatively and express their thoughts freely.

5 Evaluate ideas

After generating a plethora of ideas, you need to evaluate them and select the best ones. To do this, you can use criteria such as relevance, novelty, feasibility, and impact. You may employ rating scales, voting systems, ranking methods, or matrices to compare and prioritize your ideas. Additionally, you can group, categorize, or filter your ideas to eliminate duplicates, overlaps, or outliers. This will help you choose the most suitable option for addressing the problem or meeting the goal.

6 Implement ideas

The final step of brainstorming is to implement your selected ideas and test them in practice. You can use prototypes, experiments, simulations, or pilots to test your ideas and gather feedback. You can also use action plans, timelines, budgets, or resources to plan and execute your ideas. You should monitor and measure the results and outcomes of your ideas and make adjustments or improvements as needed.

The SCAMPER technique is another method for brainstorming and generating new ideas, using the acronym Substitute, Combine, Adapt, Modify, Put to another use, Eliminate, and Reverse to explore and innovate.

Here's a breakdown of the SCAMPER technique:

· **Substitute:**

Can you substitute any part, material, or process with something else? For example, can you use a different type of material in the manufacturing process, or another person/machine to perform a task?

Combine:

Can you combine two or more ideas or elements to create something new? For example, can you merge different technologies or processes?

Adapt:

Can you adapt the idea to suit a new situation or purpose? For example, can you take an existing product and modify it for a different market?

Modify:

Can you modify the idea by making it bigger, smaller, stronger, or more efficient? For example, can you improve a process or product by making it faster or more economical?

Put to Another Use:

Can you put the idea to a different purpose or use than its intended use? For example, can you take a product and find an innovative application in another field?

Eliminate:

Can you eliminate any unnecessary parts or features? For example, can you make the product simpler or more streamlined?

Reverse:

Can you reverse the order of actions, ingredients, or steps? For example, can you find a new way to perform a task by working backwards?

Summary of problem solving skills

Good problem solving skills are fundamentally important if you're going to be successful in your career. Certainly, problems are something that we don't particularly like. They're time-consuming. They muscle their way into already packed schedules. They force us to think about an uncertain future. And they never seem to go away! Hence the need to try and eliminate them as quickly as possible. To be an effective problem-solver, you need to be systematic and logical in your approach.

And as you increase your confidence with solving problems, you'll be less likely to rush to the first solution – which may not necessarily be the best one. It is critical, at this stage, to underline that problem-solving skills have value in many areas of life. Employers value employees who can solve problems without the need to always find a supervisor. Entrepreneurs may need to think outside the box to create a business that thrives and stands out from other similar businesses. A student who excels in problem solving can earn higher grades in some subjects like math, science and technical subjects. The skill set for problem solving in all these areas have the same components.

Recognize the Problem

First, you must recognize there is a problem before you can solve it. If something doesn't feel right or doesn't provide the expected outcome, you may have a problem. Inspect the situation and determine whether or not a problem exists.

Analyze and Investigate

You might use root cause analysis for this step. Ask why there is a problem and why things aren't occurring as expected. Continue to ask why questions until you have a good picture of the

problem. Cause and effect assessment may also provide valuable information. It may not only tell you why you have a problem, it may also provide clues necessary to solve the problem.

Brainstorm Solutions

Take the time to come up with as many practical solutions as you can. Involve your team, if a team exists. Don't discount answers at this stage. Take them all in and list them so you can explore possible solutions. Don't be afraid to think outside the box and suggest solutions that may not be part of the standard process.

Explore Options

Take your list and begin to weed out the impractical, so you can focus on the practical. If you have a team, you might suggest each team member takes one or two suggestions and investigate how that suggestion could correct the problem. Analyze the changes that would occur in the overall process, not just in the immediate problem. You don't want to create a new problem with your solution. You may find making a list of all the steps involved in your problem situation will help you identify possible consequences and missteps. You can also share solutions with other team members to see if others see potential benefits and risks you missed.

Make a Decision

Choose one or two possible solutions and implement them. You may want to implement solutions one at a time. Watch the situation and evaluate how your solution changes things. Careful observation may allow you to alter your choices quickly if the solution doesn't solve the problem or creates changes you didn't expect. Keep track of problems you successfully solve to present at job evaluations, interviews and meetings with company management. Your quick thinking and effective problem solving may help you advance in your career.